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Hybrid Multi-Task Attention-Based Deep Learning Model for Joint Prediction of Heart Disease and Diabetes Risk

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Abstract

Cardiovascular diseases (CVDs) and Type 2 diabetes mellitus (T2D) represent co-occurring metabolic and vascular conditions that account for substantial global morbidity. Although shared clinical pathways and risk factors link both conditions, conventional predictive frameworks evaluate heart disease and diabetes independently, leading to unoptimized risk stratification. Recent advancements in deep learning offer multi-task architectures capable of modelling shared representations. In this paper, a comprehensive review of hybrid multi-task attention-based deep learning paradigms for joint heart disease and diabetes risk prediction is presented. Multi-task learning (MTL) is analyzed for feature sharing, attention mechanisms are evaluated for clinical dynamic feature selection, and hybrid spatial-temporal neural networks are synthesized. Based on comparative empirical studies, critical research gaps are identified, and a novel architectural blueprint is proposed alongside future research directions.

Keywords— Solar Photovoltaic, Tilt Angle, Power Generation

I. INTRODUCTION

Cardiovascular disease and Type 2 diabetes mellitus remain two of the most prevalent chronic conditions affecting public health worldwide. Although clinically managed as separate entities, extensive pathophysiological research confirms a metabolic overlap between vascular degeneration and glycemic regulation. Biomarkers such as blood pressure, HbA1c, body mass index, and lipid panels yield prognostic signals relevant to both disease outcomes. Early identification of patient risk remains vital to effective clinical intervention; however, traditional medical prediction tools rely on isolated, single-disease risk calculators. This siloed predictive approach fails to capture complex cross-condition dependencies, frequently underestimating net cardiovascular risk in diabetic cohorts.

To overcome the shortcomings of independent predictive models, deep learning techniques have evolved toward multi-task frameworks. The core methodology incorporates three distinct design principles:

A. Task Learning (MTL): MTL models construct shared internal feature spaces to assess diabetes and heart

disease simultaneously from multi-modal inputs. Hard parameter sharing across bottom layers encourages representation sharing, mitigating overfitting while capturing non-obvious cross-disease correlations.

B. Attention Mechanism: Attention layers dynamically assign weights to time-series clinical variables according to temporal relevance. By prioritizing critical fluctuations (such as recent blood pressure variances over historic baselines), attention modules improve feature attribution and clinical interpretability.

C. Hybrid Spatial-Temporal Architecture: Fusing Convolutional Neural Networks (CNNs) with recurrent topologies (LSTMs) or Transformer models enables deep architectures to process multi-modal health records. CNNs extract localized diagnostic representations (e.g., ECG signal topologies), whereas temporal networks track longitudinal trajectories within Electronic Health Records (EHRs).

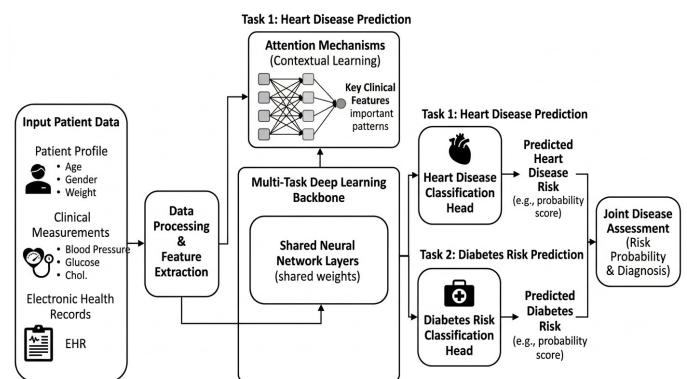


Figure 1. Graphical representation of the present study

II. LITERATURE SURVEY

Table I presents a systematic matrix of state-of-the-art literature (2022–2026) focusing on multi-task, attention-

guided, and hybrid machine learning paradigms for joint diabetes and cardiovascular risk prediction.

S. No.	Paper / Author (Year)	Condition	Model / Architecture	Dataset	Accuracy	Limitations
1	Tsai et al. (Scientific Reports, 2025)	Diabetes, Heart Disease, Stroke, Hypertension	MAND-LSTM / MAND-MHSA Multi-Task Multimodal Network	555,124 patient records (Taiwan HWDC)	AUC = 0.8918 (avg. across 4 tasks); Balanced Acc: 60%–80%	Prevalence imbalance caused higher false-negative rates for rare conditions (stroke at 8.7%).
2	Cardio Meta (arXiv, 2026)	Diabetes, Hypertension, CVD	Multi-Task Calibrated Tabular Transformer	Population survey + hospital EHR	Focuses on calibration metrics & cross-dataset transfer	Cross-dataset transferability is not yet clinically validated.
3	MMAE Framework (ScienceDirect, 2025)	T2DM + Comorbidities (incl. CVD)	Multi-Task Attention + Comorbidity Diffusion Coefficient	Real-world public healthcare dataset	Significant baseline improvement on AUC-ROC & AUC-PR	Full numerical benchmarking requires full-text access.
4	Science Direct Study (2026)	Heart Disease (5 Cardiac Tasks)	Multi-Task + Transfer Learning + 1D CNN-ResNet + Self-Attention	21,837 PTB-XL ECG + 303 clinical samples	96.8% Accuracy; ~650k parameters; 9.04 ms inference	Limited fine-tuning cohort size (303 paired clinical samples).
5	Attentive MTL (Scientific Reports, 2022)	Diabetes & Comorbidities (Diabetic CVD)	Multi-Task Deep Neural Network with Attentive Interactions	40,913 patient EHRs (Ruijin Hospital, Shanghai)	Joint Acc: 84.67%; Diabetic CVD Acc: 82.48%	High reliance on routine laboratory panels creates vulnerability to missing data.
6	Hybrid ML Approach (Biomed. Phys. Express, 2025)	CVD Risk in Diabetic Patients	Hybrid LSTM + Traditional Machine Learning	1,000 diabetic patients (Pakistan cohort)	Outperformed baseline models	Single-region cohort; records with >20% missing values excluded.
7	Kahraman et al. (2025)	Heart Disease & Comorbidities	Systematic Analytical Review & Gradient Boosting (CatBoost)	311,710 integrated patient records	CatBoost achieved top AUC up to 94.1%	Data alignment challenges across heterogeneous multi-source systems.
8	DNHRV Framework (Tandfonline, 2023)	CVD Risk in Type-2 Diabetic Males	DNHRV: DNN + DCNN Hybrid using HRV & Clinical Data	HRV + imaging + clinical history	Reported high predictive performance	Male-only cohort; heavily dependent on clean HRV signal quality.

S. No.	Paper / Author (Year)	Condition	Model / Architecture	Dataset	Accuracy	Limitations
9	Multimodal MTL Network (Scientific Reports, 2025)	Heart Disease, Diabetes, Stroke, Hypertension	Multimodal Multi-Task Attention Network	10-year nationwide longitudinal EHR database	Joint Multi-Disease AUC: 87.20%; Heart Disease AUC: 88.60%	High computational and memory complexity for long temporal sequence processing.
10	Xie et al. (2025)	Diabetes Risk	Gradient Boosting & Supervised Ensemble Learning	100,000 EHR & lifestyle survey records	Gradient Boosting AUC: 89.40% (Baseline LR: 78.20%)	Performance drops when continuous biochemical panel inputs are incomplete.

The structured literature matrix reflects a distinct trend toward combining spatial, temporal, and attention mechanisms for multi-disease risk evaluation. Tsai et al. (2025) demonstrated the feasibility of processing nationwide electronic health records (555,124 patients) using MAND-LSTM and multi-head self-attention, obtaining an average AUC of 0.8918 across four chronic conditions. However, performance variations were observed in low-prevalence targets like stroke (8.7% prevalence), indicating that multi-task models remain sensitive to target label distribution. In parallel, Cardio Meta (2026) shifted focus toward tabular transformers and calibration performance, prioritizing risk score reliability across hospital system over raw accuracy metrics. The MMAE framework (2025) integrated a comorbidity diffusion coefficient alongside task attention, yielding improvements in precision-recall curves for T2DM patients. Lightweight architectures were explored in the ScienceDirect 2026 study, where a 1D CNN-ResNet paired with transfer learning attained 96.8% accuracy and 9.04 ms inference time on 21,837 ECG records, albeit constrained by a small clinical validation cohort (303 patients). Studies focused on specific comorbidities, such as the Attentive MTL model on Ruijin Hospital records (40,913 patients), achieved 84.67% joint accuracy and 82.48% accuracy for diabetic CVD. Hybrid models incorporating physiological signals (e.g., DNHRV combining HRV, imaging, and clinical metrics) confirmed the benefit of multi-modal features, though signal noise sensitivity and single-gender cohorts restrict general applicability. Finally, ensemble algorithms benchmarked by Kahraman et al. (2025) (CatBoost AUC up to 94.1% on 311,710 records) and Xie et al. (2025) (Gradient Boosting AUC 89.40% on 100,000 records) underscore the necessity of robust data harmonization when unifying heterogeneous healthcare systems.

IV. COMPARATIVE ANALYSIS

Comparative examination of existing methodologies reveals key trade-offs across model complexity, data representations, and predictive performance:

7 A. Single-Task vs. Multi-Task Learning: Single-task models (such as standard Logistic Regression or isolated Gradient Boosting trees) achieve strong benchmark accuracy on homogeneous single-disease datasets. However, they lack the capacity to transfer cross-domain representations. Multi-task neural frameworks share lower-layer parameters, effectively capturing shared metabolic pathways and yielding consistent performance gains across concurrent risk evaluations.

B. Tabular Ensembles vs. Deep Spatial-Temporal Architectures: Gradient boosting algorithms (e.g., CatBoost, XGBoost) demonstrate superior handling of static, tabular clinical panels with low preprocessing overhead. Conversely, hybrid deep learning architectures (CNN-LSTM / Transformers) excel at processing complex, unstructured, and continuous time-series trajectories (e.g., continuous glucose monitoring, ECG waveforms, and multi-year EHR sequences), despite higher computational demand.

C. Attention Mechanisms vs. Static Feature Selection: Traditional feature selection techniques rely on static linear correlations. Integrating self-attention and dynamic attentive modules allows models to weight temporal clinical indicators contextually, enhancing interpretability while maintaining high discriminatory power (AUC > 0.88 across multiple benchmark studies).

V. RESEARCH GAPS

Despite recent progress, critical challenges persist in current literature:

1 A. High Missingness in Clinical Records: Existing models exhibit significant degradation when continuous biochemical inputs (e.g., HbA1c panels, lipid profiles) contain missing entries, restricting deployment in resource-constrained primary care settings.

B. Dataset Imbalance and Generalizability: Class imbalance between common metabolic conditions and lower-prevalence cardiovascular events frequently leads to inflated false-negative rates for critical complications.

C. Cross-Institutional Validation: Most deep multi-task architectures are evaluated on localized hospital registries or single-region cohorts, lacking external validation across diverse demographic populations and multi-site healthcare protocols.

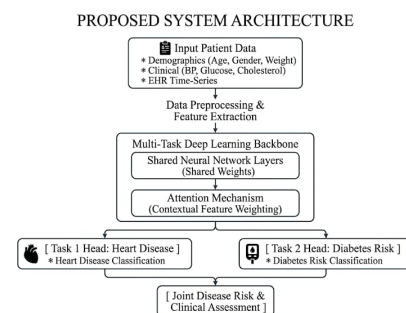
D. Computational Overhead of Temporal Sequences: Processing multi-year longitudinal EHR embeddings using dense self-attention mechanisms introduces significant computational and memory complexity, hindering real-time clinical decision support.

VI. PROPOSED FRAMEWORK / FUTURE DIRECTION

To address the identified gaps, a comprehensive hybrid multi-task attention-guided deep learning framework is proposed. The system architecture is depicted in Fig. 1.

Interpretability:- High-performing deep learning models frequently lack intrinsic explainability. Clinicians require clear rationale for risk predictions, yet few joint models incorporate internal self-attention visualization or unified SHAP-based interpretations to explain multi-disease predictions.

Practical Deployment Obstacles: Current studies rarely address operational integration into clinical decision support systems (CDSS). Issues such as handling missing lab values in real time, balancing multi-task loss functions during streaming updates, and maintaining inference speed in resource-constrained medical environments are under-researched.



Future Research Directions:

1) Federated Learning Integration: Decentralized privacy-preserving training protocols should be integrated to enable cross-institutional model training without centralizing sensitive patient EHR data.

2) Robust Imputation Strategies: Graph-based and generative adversarial imputation techniques should be incorporated to handle missing clinical panel values effectively.

3) Lightweight Attention Mechanisms: Linearized self-attention modules should be explored to reduce memory overhead during long-term temporal sequence processing.

VII. CONCLUSION

This paper provided a structured review of hybrid multi-task attention-based deep learning architectures for the joint prediction of heart disease and diabetes risk. By moving beyond isolated single-disease models, multi-task architectures leverage shared biological pathways to improve overall risk stratification and diagnostic generalizability. Attention mechanisms further enhance model interpretability by isolating key temporal clinical indicators. Addressing

challenges related to data missingness, cross-institutional validation, and

privacy via federated learning will pave the way for scalable, real-time clinical decision support systems

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